Use Case Flow - Smart Complaint Management System

Use Case ID	UC-01
Use Case Name	Submit Complaint
Actor	Citizen
Precondition	Citizen must be logged in.
Main Flow	1. Citizen taps "New Complaint". 2. Fills in title, description, and severity. 3. System suggests top-3 categories via NLP. 4. Citizen accepts or overrides the suggested category. 5. Attaches images and enables GPS (optional). 6. Citizen taps "Submit". 7. System validates all mandatory fields. 8. System generates a unique ticket ID. 9. System routes the complaint to the appropriate department/agent. 10. System sends a confirmation notification with the ticket ID.
Alternative Flow	7a. If mandatory fields are missing, system displays validation errors and the citizen corrects them. **Rejoins main flow at step 6.**
Postcondition	Complaint ticket is created, stored, and routed to the appropriate department.

Use Case ID	UC-02
Use Case Name	Track Complaint Status
Actor	Citizen
Precondition	Citizen must be logged in and have at least one submitted complaint.
Main Flow	1. Citizen navigates to "My Complaints". 2. Selects a complaint from the list. 3. System displays current status and timeline. 4. Citizen views status history with timestamps.
Alternative Flow	2a. If the selected complaint is invalid or deleted, system displays an error and the citizen selects another complaint. **Rejoins main flow at step 2.**
Postcondition	Citizen views the up-to-date status of the selected complaint.

Use Case ID	UC-03
Use Case Name	Rate Resolution
Actor	Citizen
Precondition	Complaint status must be "Resolved".
Main Flow	1. Citizen opens the resolved complaint. 2. Selects a star rating (1-5). 3. Enters an optional comment. 4. Submits the feedback. 5. System saves the rating and updates the CSAT score.
Alternative Flow	4a. If rating submission fails, system displays an error and prompts the citizen to retry. **Rejoins main flow at step 4.**
Postcondition	Rating and feedback are recorded; complaint moves to "Closed".

Use Case ID	UC-04
Use Case Name	Resolve Complaint
Actor	Agent
Precondition	Complaint must be assigned to the agent.
Main Flow	1. Agent opens the assigned complaint. 2. Reviews complaint details, attachments, and notes. 3. Performs the required action (field visit, coordination, fix). 4. Adds resolution notes describing the action taken. 5. Marks the complaint as "Resolved". 6. System notifies the citizen.
Alternative Flow	3a. If more information is needed, agent sets status to "Pending Info" and queries the citizen; the citizen replies with the details. **Rejoins main flow at step 2.**
Postcondition	Complaint is marked "Resolved" and the citizen is notified.

Use Case ID	UC-05
Use Case Name	Add Notes and Evidence
Actor	Agent
Precondition	Agent must be assigned to the complaint.
Main Flow	1. Agent opens the complaint. 2. Selects "Add Note". 3. Enters note text describing progress or findings. 4. Attaches evidence file (optional, PDF/image, max 10 MB). 5. Saves the note.
Alternative Flow	4a. If the attachment exceeds 10 MB, system rejects the file and the agent attaches a smaller file. **Rejoins main flow at step 4.**
Postcondition	Note and evidence are stored in the complaint's resolution trail.

Use Case ID	UC-06
Use Case Name	Escalate Complaint
Actor	Supervisor
Precondition	Complaint must be stalled beyond SLA or flagged for review.
Main Flow	1. Supervisor opens the "Overdue" tab. 2. Selects the stalled complaint. 3. Clicks "Escalate". 4. Enters an escalation reason and selects the senior team. 5. System moves the ticket to the priority queue and resets the SLA timer. 6. System notifies the original agent, new team, and citizen.
Alternative Flow	4a. If no senior team is configured for the category, system alerts the admin and the supervisor assigns a senior agent manually. **Rejoins main flow at step 5.**
Postcondition	Complaint is reassigned to the priority queue with a new SLA; all parties are notified.

Use Case ID	UC-07
Use Case Name	Auto-Categorise Complaint
Actor	System (invoked by Citizen/Agent)
Precondition	Complaint description text must be available.
Main Flow	1. System receives the complaint description text. 2. NLP model processes the text and ranks the top-3 categories. 3. System returns confidence scores for each suggestion. 4. Citizen or agent accepts or overrides the suggestion.
Alternative Flow	3a. If all confidence scores are below 70%, system shows the full category list instead of suggestions. **Rejoins main flow at step 4.**
Postcondition	Complaint is tagged with a category before routing.

Use Case ID	UC-08
Use Case Name	View Dashboard
Actor	Supervisor, Admin
Precondition	User must have Supervisor or Admin role.
Main Flow	1. User navigates to "Dashboard". 2. System loads KPIs (open, resolved, overdue, avg. resolution time). 3. User applies filters (date range, department, agent). 4. System refreshes the displayed data every 60 seconds.
Alternative Flow	3a. If no data matches the selected filters, system displays an empty-state message and the user changes the filters. **Rejoins main flow at step 3.**
Postcondition	User views up-to-date performance metrics for the selected filters.

Use Case ID	UC-09
Use Case Name	Manage Users
Actor	Admin
Precondition	Admin must be logged in.
Main Flow	1. Admin navigates to "User Management". 2. Clicks "Add User". 3. Enters name, email, department, and role. 4. Saves the account. 5. System sends a welcome email to the new user.
Alternative Flow	3a. If the email already exists, system displays a duplicate-account error and the admin enters a different email. **Rejoins main flow at step 3.**
Postcondition	New user account is created with the correct role permissions.

Use Case ID	UC-10
Use Case Name	Configure SLA Rules
Actor	Admin
Precondition	Admin must be logged in.
Main Flow	1. Admin navigates to "SLA Configuration". 2. Selects a complaint category and severity level. 3. Sets response-time and resolution-time targets. 4. Saves the rule.
Alternative Flow	3a. If entered values are invalid (e.g. negative time), system rejects them and the admin corrects the values. **Rejoins main flow at step 3.**
Postcondition	SLA rule is updated and applied to new complaints immediately.

Use Case ID	UC-11
Use Case Name	Log In and Authenticate
Actor	Citizen, Agent, Supervisor, Admin
Precondition	User must have a registered account.
Main Flow	1. User opens the login page. 2. Enters email and password (or phone number for OTP). 3. System verifies the credentials. 4. System issues a session token containing the user's role. 5. System redirects the user to the home screen for their role. 6. System records the login in the audit log.
Alternative Flow	3a. If the credentials are invalid, system displays an error and the user re-enters them. **Rejoins main flow at step 2.**
Postcondition	User is logged in and can access only the features permitted for their role.

Use Case ID	UC-12
Use Case Name	Monitor SLA and Send Alerts
Actor	System (timer)
Precondition	SLA rules are configured and the complaint is open.
Main Flow	1. System checks every open complaint's SLA deadline at a fixed interval. 2. System detects a complaint with less than 2 hours remaining. 3. System sends a warning alert to the assigned agent (in-app and email). 4. System detects a complaint whose deadline has passed. 5. System sends a breach alert to the supervisor and flags the complaint as overdue. 6. System records each alert in the complaint's audit log.
Alternative Flow	3a. If the email service is unavailable, system delivers the alert in-app only and queues the email for retry. **Rejoins main flow at step 4.**
Postcondition	Assigned staff are alerted before and at SLA breach, and every alert is logged.